

Solution Architecture

Introduction

Solution Architecture describes the overall structure of the AI Learning Assistant and the interaction between different modules.

Architecture Components

User

Provides study-related text input.

Streamlit Interface

Receives user input and displays results.

Emotion Detection Module

Detects the emotional state.

Gemini AI

Generates personalized learning guidance.

Logger

Stores interaction details.

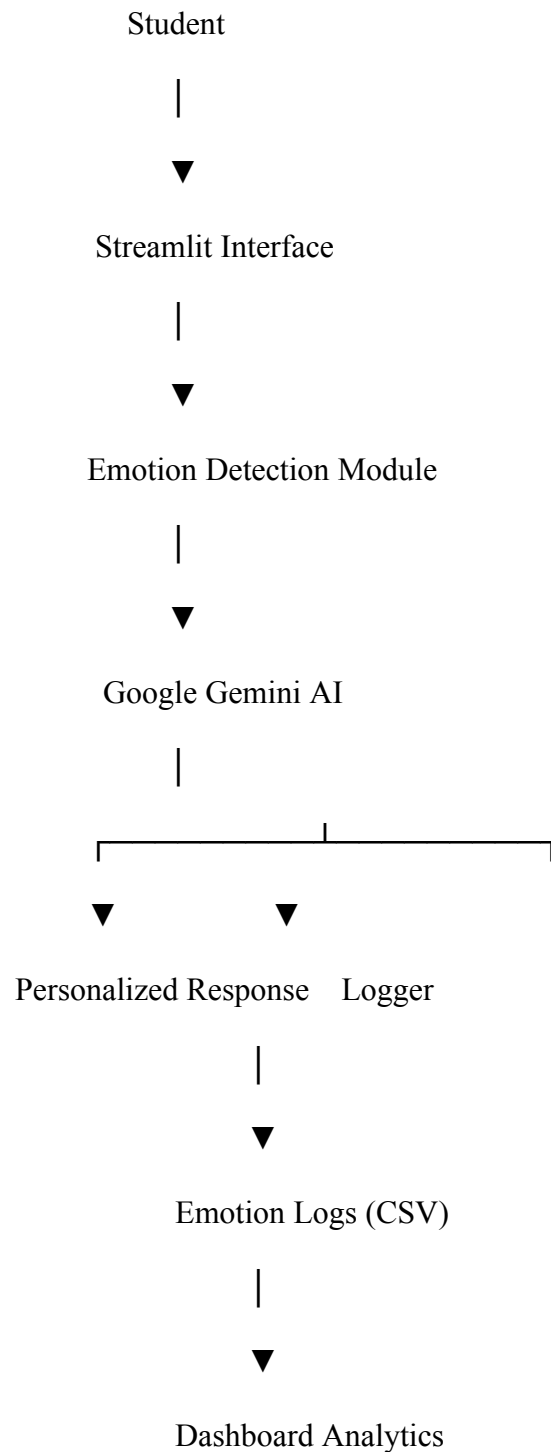
CSV Database

Maintains user interaction records.

Dashboard

Displays analytics and reports.

Architecture Flow



Advantages

- Modular architecture
- Easy maintenance
- Scalable
- Secure

- Fast processing
 - Supports future enhancements
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Future Enhancements

- Voice emotion detection
 - Facial emotion recognition
 - Mobile application
 - Database integration (MySQL/Firebase)
 - Multilingual support
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Conclusion

The solution architecture provides a modular and scalable design that enables efficient interaction between the user interface, emotion detection, Gemini AI, logging system, and analytics dashboard.